ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)

path to success

PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Kinematics -1D & Calculus

(English Medium)

EXERCISE (O)

SINGLE CORRECT TYPE QUESTIONS

Rectilinear motion with constant acceleration

1. An athlete completes one round of a circular track of radius R in 40 sec. What will be his displacement at the end of 2 min. 20 sec

(A) Zero (B) 2R (C) 2πR (D) 7πR

KM0287

2. A particle moves in straight line in same direction for 20 seconds with velocity 3 m/s and then moves with velocity 4 m/s for another 20 sec and finally moves with velocity 5 m/s for next 20 seconds. What is the average velocity of the particle?

(A) 3 m/s (B) 4 m/s (C) 5 m/s (D) Zero

KM0288

3. A body covers first $\frac{1}{3}$ part of its journey with a velocity of 2 m/s, next $\frac{1}{3}$ part with a velocity of 3 m/s and rest of the journey with a velocity 6m/s. The average velocity of the body will be

(A) 3 m/s (B) $\frac{11}{3}$ m/s (C) $\frac{8}{3}$ m/s (D) $\frac{4}{3}$ m/s

KM0289

4. A stone is thrown vertically upward with an initial velocity u from the top of a tower, reaches the ground with a velocity 3u. The height of the tower is :

(A) $\frac{3u^2}{g}$ (B) $\frac{4u^2}{g}$ (C) $\frac{6u^2}{g}$ (D) $\frac{9u^2}{g}$

KM0290

5. A body moving on straight line such that at t = 0, u = 5 m/s and acceleration a = –1 m/s^2, speed at t =10 s is

(A) –5 m/s (B) 5 m/s (C) 10 m/s (D) –10 m/s

KM0291

6. The average velocity of a freely falling body is numerically equal to half of the acceleration due to gravity. The velocity of the body as it reaches the ground is

(A) g (B) $\frac{g}{2}$ (C) $\frac{g}{\sqrt{2}}$ (D) $\sqrt{2}g$

KM0292

7. A particle is dropped from certain height. The time taken by it to fall through successive distances of 1 m each will be:

(A) All equal, being equal to 2 / g second.

(B) In the ratio of the square roots of the integers 1,2, 3, ...

(C)) In the ratio of the difference in the square roots of the integers, i.e.,

$\sqrt{1},(\sqrt{2}-\sqrt{1}),(\sqrt{3}-\sqrt{2}),(\sqrt{4}-\sqrt{3})$, ...

(D) In the ratio of the reciprocals of the square roots of the integers, i.e.,

$\frac{1}{\sqrt{1}},\frac{1}{\sqrt{2}},\frac{1}{\sqrt{3}}$,.....

KM0293

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8. A ball is thrown straight upward with a speed v from a point h meter above the ground. The time taken for the ball to strike the ground is:

(A) $\frac{v}{g}\left[1+\sqrt{1+\frac{2hg}{v^2}}\right]$ (B) $\frac{v}{g}\left[1-\sqrt{1-\frac{2hg}{v^2}}\right]$ (C) $\frac{v}{g}\left[1-\sqrt{1+\frac{2hg}{v^2}}\right]$ (D) $\frac{v}{g}\left[2+\sqrt{1+\frac{2hg}{v^2}}\right]$

KM0294

9. A balloon is going upwards with velocity 12 m/sec. It releases a packet when it is at a height of 65 m from the ground. How much time the packet will take to reach the ground ($g = 10\ m/sec^2$)

(A) 5 sec (B) 6 sec (C) 7 sec (D) 8 sec

KM0295

10. A bullet moving at 20 m/sec. It strikes a wooden plank and penetrates 4 cm before coming to stop. The time taken to stop is:

(A) 0.008 sec (B) 0.016 sec (C) 0.004 sec (D) 0.002 sec

KM0296

11. A body starts from rest and is uniformly accelerated for 30 s. The distance travelled in the first 10 s is x_1, next 10 s is x_2 and the last 10 s is x_3. Then $x_1 : x_2 : x_3$ is the same as :-

(A) 1 : 2 : 4 (B) 1 : 2 : 5 (C) 1 : 3 : 5 (D) 1 : 3 : 9

KM0050

12. If a body starts from rest and travels 120 cm in the 6th second, with constant acceleration then what is the acceleration :

(A) $0.20\ m/s^2$ (B) $0.027\ m/s^2$ (C) $0.218\ m/s^2$ (D) $0.03\ m/s^2$

KM0336

13. A particle has an initial velocity of 9 m/s due east and a constant acceleration of $2\ m/s^2$ due west. The distance covered by the particle in the fifth second of its motion is :-

(A) 0 (B) 0.5 m (C) 2 m (D) none of these

KM0337

14. A body falls freely from rest. It covers as much distance in the last second of its motion as covered in the first three seconds. The body has fallen for a time of :

(A) 3 s (B) 5 s (C) 7 s (D) 9 s **KM0338**

15. A parachutist jumps out of an airplane and accelerates with gravity for 6 seconds. He then pulls the parachute cord and after a 4 s deceleration period, descends at 10 m/s for 60 seconds, reaching the ground. From what height did the parachutist jump? Assume acceleration due to gravity to be $10\ m/s^2$ throughout the motion.

(A) 840 m (B) 920 m (C) 980 m (D) 1020 m **KM0339**

Motion with variable acceleration & calculus

16. If $s = 2t^3 + 3t^2 + 2t + 8$ then the time at which acceleration is zero, is :-

(A) $t = \frac{1}{2}$ (B) $t = 2$ (C) $t = \frac{1}{2\sqrt{2}}$ (D) Never

KM0340

17. For a particle moving in a straight line the position of the particle at time (t) is given by $x = \frac{t^3}{6} - t^2 - 9t + 18$ m. What is the velocity of the particle when its acceleration is zero :-

(A) 18 m/s (B) –9 m/s (C) –11 m/s (D) 6 m/s **KM0341**

18. A particle moves along a straight line such that at time t its displacement from a fixed point O on the line is $3t^2 - 2$. The velocity of the particle when t = 2 is:

(A) $8\ ms^{-1}$ (B) $4\ ms^{-1}$ (C) $12\ ms^{-1}$ (D) 0 **KM0342**

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19. Temperature of a body varies with time as $T = (T_0 + \alpha t^2 + \beta \sin t)K$, where T_0 is the temperature in Kelvin at t = 0 sec. & $\alpha = 2/\pi$. K/s^2 & $\beta = -4$ K, then rate of change of temperature at $t = \pi$sec. is

(A) 8 K (B) 8^0K (C) 8K/sec (D) 8^0K/sec **KM0343**

20. The velocity of a particle moving on the x-axis is given by $v = x^2 + x$ where v is in m/s and x is in m. Find its acceleration in m/s^2 when passing through the point x = 2m

(A) 0 (B) 5 (C) 11 (D) 30 **KM0344**

21. Velocity of a particle varies with time as v = 4t. The displacement of particle between t = 2 to t = 4 sec, is :

(A) 12 m (B) 36 m (C) 24 m (D) 6 m **KM0345**

22. A point mass moves with velocity $v = (5t - t^2)$ ms^{-1} in a straight line. Find the distance travelled (i.e. $\int v dt$) in fourth second.

(A) $\frac{31}{6}$m (B) $\frac{29}{6}$m (C) $\frac{37}{6}$m (D) None of these

KM0065

23. A particle is projected with velocity v_0 along x–axis. The deceleration on the particle is proportional to the square of the distance from the origin i.e., $a = -\alpha x^2$. The distance at which the particle stops is:–

(A) $\sqrt{\frac{3v_0}{2\alpha}}$ (B) $\left(\frac{3v_0}{2\alpha}\right)^{\frac{1}{3}}$ (C) $\sqrt{\frac{3v_0^2}{2\alpha}}$ (D) $\left(\frac{3v_0^2}{2\alpha}\right)^{\frac{1}{3}}$ **KM0066**

24. The acceleration vector along x–axis of a particle having initial speed v_0 changes with distance as $a = \sqrt{x}$. The distance covered by the particle, when its speed becomes twice that of initial speed is:–

(A) $\left(\frac{9}{4}v_0\right)^{\frac{4}{3}}$ (B) $\left(\frac{3}{2}v_0\right)^{\frac{4}{3}}$ (C) $\left(\frac{2}{3}v_0\right)^{\frac{4}{3}}$ (D) $2v_0$ **KM0067**

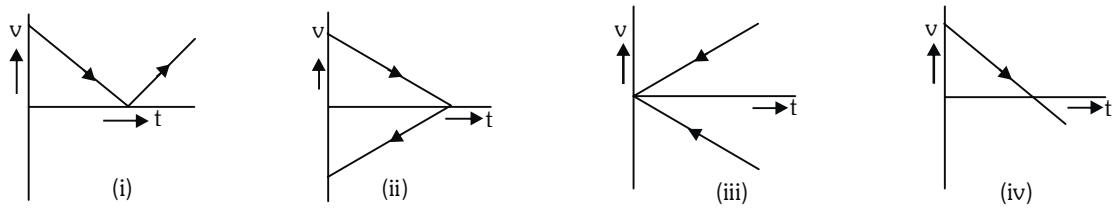
25. If x denotes displacement in time t and x = a cos t, then the acceleration is–

(A) a cos t (B) –a cos t (C) a sin t (D) –a sin t

KM0297

Questions based on graph

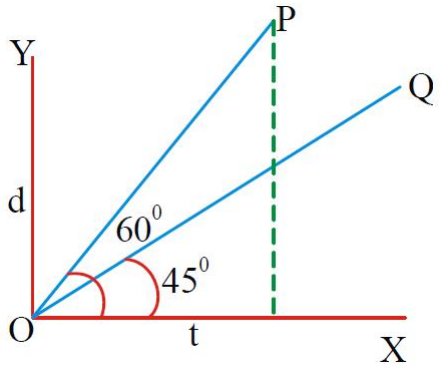
26. A ball is thrown vertically upwards. Which of the following graph/graphs represent velocity-time graph of the ball during its flight (air resistance is neglected) **KM0298**



(A) i (B) ii (C) iii (D) iv

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27. The displacement - time graphs of two bodies A and B are OP and OQ respectively. If ∠POX is 60° and ∠QOX is 45°, the ratio of the velocity of A to that of B is:



(A) $\sqrt{3}:\sqrt{2}$ (B) $\sqrt{3}:1$ (C) $1:\sqrt{3}$ (D) $3:1$

KM0299

28. Choose the correct statement :

(A) The area of displacement - time graph gives velocity.

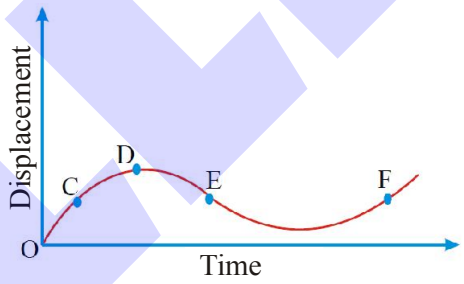
(B) The slope of velocity - time graph gives acceleration.

(C) The slope of displacement - time graph gives acceleration.

(D) The area of velocity - time graph gives average velocity.

KM0300

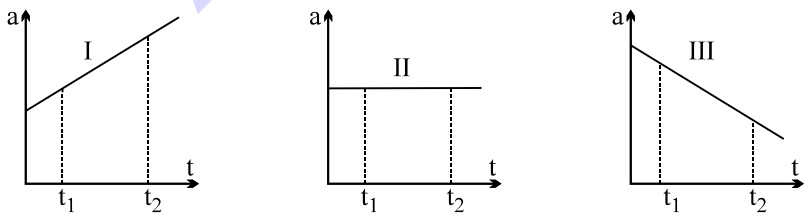
29. The displacement-time graph of a moving particle is shown in Fig. The instantaneous velocity of the particle is negative at the point



(A) D (B) F (C) C (D) E

KM0301

30. Each of the three graphs represents acceleration versus time for an object that already has a positive velocity at time t_1. Which graphs show an object whose speed is increasing for the entire time interval between t_1 and t_2?

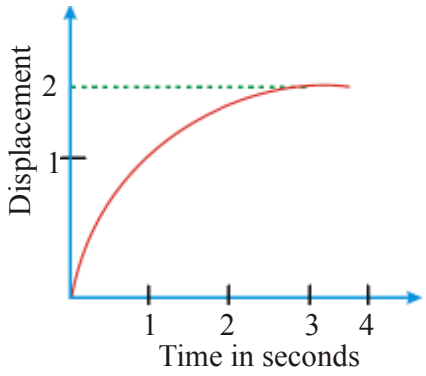


(A) graph I, only (B) graphs I and II, only

(C) graphs I and III, only (D) graphs I, II, and III **KM0349**

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31. The displacement of a particle as a function of time is shown in the figure. The figure shows that



(A) the particle starts with certain velocity but the motion is retarded and finally the particle stops.

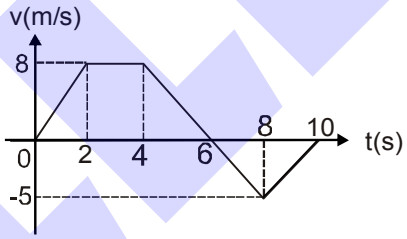
(B) the velocity of the particle is constant through out.

(C) the acceleration of the particle is constant throughout.

(D) the particle starts with constant velocity, then motion is accelerated and finally the particle moves with another constant velocity.

KM0302

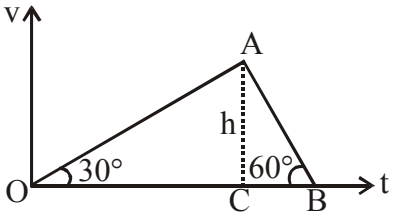
32. For a particle moving along x-axis, velocity-time graph is as shown in figure. Find the distance travelled and displacement of the particle?



(A) Displacement = 42m & distance = 22m
(B) Displacement = Distance = 42m
(C) Displacement = Distance = 22m
(D) Displacement = 22 m & Distance = 42m

KM0303

33. The velocity time (v – t) graph of a body is shown in figure. For the intervals OC & CB, the ratio of the distances covered is :-

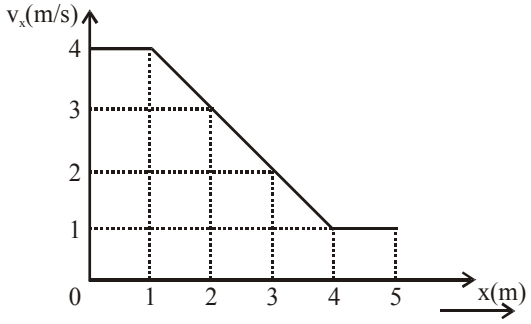


(A) 3 : 1
(B) 1 : 3
(C) $\sqrt{3}:1$
(D) $\sqrt{3}:2$

KM0304

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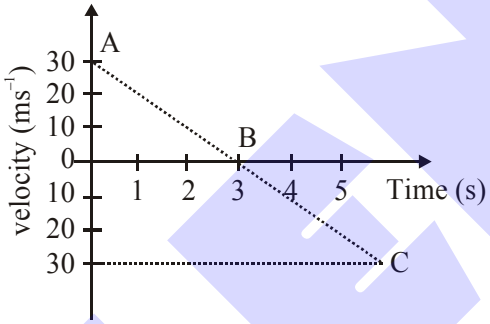
34. A body moves in a straight line along x-direction for which graph of v_x vs x is shown in the graph. Find the magnitude of acceleration of the body at x = 3m?



(A) 2 (B) 0 (C) 4 (D) –2

KM0305

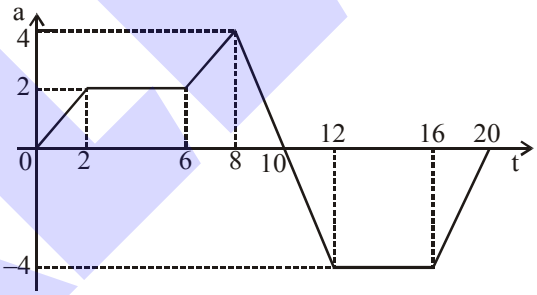
35. The velocity-time graph of a stone thrown vertically upward with an initial velocity of 30ms^{-1} is shown in the figure. The velocity in the upward direction is taken as positive and that in the downward direction as negative. What is the maximum height to which the stone rises?



(A) 30 m (B) 45 m (C) 60 m (D) 90 m

KM0306

36. If initial velocity of particle is 2 m/s, the maximum velocity of particle from t = 0 to t = 20 sec is :

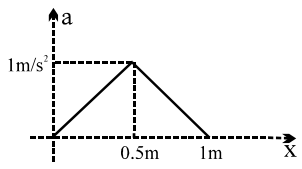


(A) 20 m/s (B) 18 m/s (C) 22 m/s (D) 24 m/s **KM0354**

37. The engine of a motorcycle can produce a maximum acceleration 5 m/s^2. Its brakes can produce a maximum retardation 10 m/s^2. If motorcyclist start from point A and reach at point B. What is the minimum time in which it can cover if distance between A and B is 1.5 km. (Given : that motorcycle comes to rest at B)

(A) 30 sec (B) 15 sec (C) 10 sec (D) 5 sec **KM0355**

38. A body initially at rest, starts moving along x-axis in such a way so that its acceleration vs displacement plot is as shown in figure. The maximum velocity of particle is :-



(A) 1 m/s (B) 6 m/s (C) 2 m/s (D) none **KM0356**

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MULTIPLE CORRECT TYPE QUESTIONS

39. A particle moving along a straight line with uniform acceleration has velocities 7m/s at A and 17 m/s at C. B is the mid point of AC. Then
(A) The velocity at B is 12 m/s.
(B) The average velocity between A and B is 10 m/s.
(C) The ratio of the time to go from A to B to that from B to C is 3 : 2.
(D) The average velocity between B and C is 15 m/s. **KM0357**

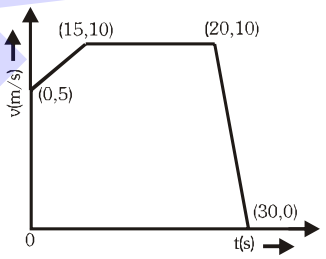
40. A particle moves along the X–axis as $x = u(t-2s) + a(t-2s)^2$
(A) The initial velocity of the particle is u (B) The acceleration of the particle is a
(C) The acceleration of the particle is 2a (D) At t = 2s particle is at the origin. **KM0092**

41. A ball is dropped from a building. Somewhere down it crosses a window of length 4 m in 0.5 sec. Speed of ball at top of window is v_1 and at bottom v_2, then choose the **CORRECT** option(s) ($g = 10\ m/s^2$) :-

(A) $v_2 - v_1 = 5$ m/s (B) $v_2 + v_1 = 16$ m/s (C) $\frac{v_2}{v_1} = 9$ (D) $\frac{v_2}{v_1} = \frac{21}{11}$

KM0358

42. The figure shows the velocity time graph of a particle which moves along a straight line starting with velocity at 5 m/sec and coming to rest at t = 30s . Then :



(A) Distance travelled by the particle is 212.5 m
(B) Distance covered by the particle when it moves with constant velocity is 100 m
(C) Velocity of the particle at t = 25s is 5 m/sec
(D) Velocity of the particle at t = 9s is 8 m/sec.

KM0307

43. The motion of a body is given by the equation $\frac{dv}{dt} = 6.0 - 3v$, where v is speed in m s^{-1} and *t* is in second. If body was at rest at t = 0, then
(A) the speed varies with the time as $v = 2(1 - e^{-3t})$ m s^{-1}
(B) the terminal speed is 1.0 m s^{-1}
(C) the magnitude of the initial acceleration is 2.0 m s^{-2}
(D) the speed is 1 m s^{-1} when the acceleration is half the initial value.

KM0308

44. If velocity of the particle is given by $v = \sqrt{x}$, where x denotes the position of the particle and initially particle was at x = 4m, then which of the following are correct.
(A) At t = 2 s, the position of the particle is at x = 9m
(B) Particle acceleration at t = 2 s is 1 m/s^2
(C) Particle acceleration is 1/2 m/s^2 through out the motion
(D) Particle will never go in negative direction from its starting position

KM0309

45. A particle moves with an initial velocity v_0 and retardation βv, where v is its velocity at any time t (β is a positive constant).
(A) the particle will cover a total distance of v_0/β.
(B) the particle will continue to move for a very long time.
(C) the particle will stop shortly.
(D) the velocity of particle will become $v_0/2$ after time $1/\beta$.

KM0310

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COMPREHENSION TYPE QUESTIONS

Paragraph for Question no. 46 and 47

A particle is moving in a straight line along positive y-axis. Its displacement from origin at any time t is given by $y = 5t^2 - 10t + 5$ where y is in meters and t is in seconds.

46. The velocity at t = 2s will be :

(A) 20 ms^{-1} (B) 10 ms^{-1} (C) 5 ms^{-1} (D) 15 ms^{-1}

KM0363

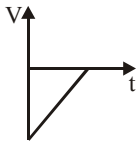
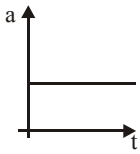
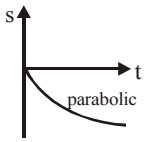
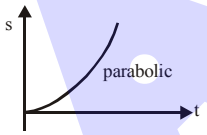
47. Displacement of particle when its velocity is zero, is

(A) 2.5 m (B) 1.25 m (C) 5m (D) 0m

KM0080

MATRIX MATCH TYPE QUESTION

48. v, a, s and t denote velocity, acceleration, displacement and time respectively. Match the columns :-

Column-I		**Column-II**	
(A)		(P)	Velocity of the particle is in positive direction, acceleration in negative direction
(B)		(Q)	Both velocity and acceleration of the particle are in negative directions.
(C)		(R)	Velocity of the particle is in negative direction and acceleration in positive direction
(D)		(S)	Velocity and acceleration both in positive direction
		(T)	Acceleration is constant

KM0081

49. A balloon starts rising up from ground with constant net acceleration of $10 m/s^2$. After 2 s a particle drops from the balloon. After further 2 s match the following : ($g = 10 m/s^2$)

Column-I		**Column-II**	
(A)	Height of particle from ground	(P)	Zero
(B)	Speed of particle	(Q)	10 SI units
(C)	Displacement of Particle	(R)	40 SI units
(D)	Acceleration of particle	(S)	20 SI units

KM0364

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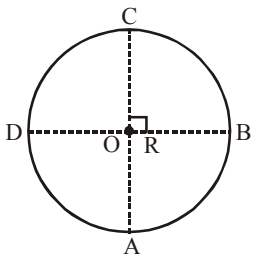
EXERCISE (S)

Motion with constant acceleration

1. A car travelling at 72 km/h decelerates uniformly at 2 m/s^2. Calculate (a) the distance it goes before it stops, (b) the time it takes to stop, and (c) the distance it travels during the first and third seconds.

KM0311

2. A particles starts from point A with constant speed v on a circle of radius R. Find magnitude of average velocity during its journey from :-



(a) A to B (b) A to C (c) A to D **KM0001**

3. A driver travelling at speed 36 kmh^{-1} sees the light turn red at the intersection. If his reaction time is 0.6s, and then the car can deaccelerate at 4ms^{-2}. Find the stopping distance of the car.

KM0007

4. From the top of a tower, a ball is thrown vertically upwards. When the ball reaches h below the tower, its speed is double of what it was at height h above the tower. Find the greatest height attained by the ball from the tower.

KM0014

5. A balloon rises from rest on the ground with constant acceleration $\frac{g}{3}$. A stone is dropped when the balloon has rises to a height 60 metre. The time taken by the stone to reach the ground is.

KM0016

Motion with variable acceleration & calculus

6. The position x of a particle w.r.t. time t along x–axis is given by $x = 9t^2 - t^3$ where x is in metre and t in second. Find
(a) Maximum speed along +x direction
(b) Position of turning point
(c) Displacement in first ten seconds
(d) Distance travelled in first ten seconds **KM0017**

7. The momentum of a particle moving in straight line is given by $p = \ln t + \frac{1}{t}$ (in kg m/s) find the time $t > 0$ at which the net force acting on particle is 0 and it's momentum at that time. [Hint : $F = \frac{dp}{dt}$]

KM0018

8. The velocity of the particle is given as $v = 3t^3 + t - \frac{1}{t^2}$. Calculate the net force acting on the body at time t = 2 sec, if the mass of the body is 5 kg.

KM0019

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9. The charge flowing through a conductor beginning with time t = 0 is given by the formula $q = 2t^2 + 3t + 1$ (coulombs). Find the current $i = \frac{dq}{dt}$ at the end of the 5th second.

KM0021

10. The motion of a particle in a straight line is defined by the relation $x = t^4 - 12t^2 - 40$ where *x* is in meters and t is in sec. Determine the position x, velocity v and acceleration a of the particle at t = 2sec.

KM0023

11. A point moves in a straight line so that its distance from the start in time t is equal to

$$s = \frac{1}{4}t^4 - 4t^3 + 16t^2.$$

(a) At what times was the point at its starting position?
(b) At what times is its velocity equal to zero? **KM0024**

12. A body whose mass is 3 kg performs rectilinear motion according to the formula $s = 1 + t + t^2$, where *s* is measured in centimetres & t in seconds. Determine the kinetic energy $\frac{1}{2}mv^2$ of the body in 5 sec after its start.

KM0025

13. A force of 40N is responsible for the motion of a body governed by the equation $s = 2t + 2t^2$ where s is in meters and t in sec. What is the momentum of the body at t = 2 sec?
[**Hint**: Find acc. then m = F/a & p = mv] **KM0026**

14. The acceleration of a particle starting from rest vary with respect to time is given by a = (2t – 6), where t is in seconds. Find the time (in seconds) at which velocity of particle in negative direction is maximum.

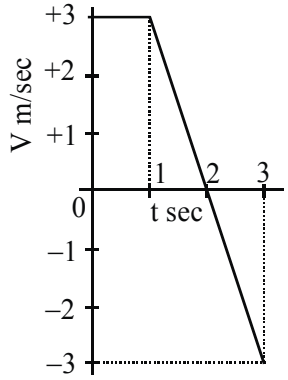
KM0028

15. Position vector of a particle is given by $\vec{r} = 3t^3\hat{i} + 4t\hat{j} + t^2\hat{k}$ .find avg acceleration of particle from t = 1 to t = 2sec.

KM0030

Question based on graph

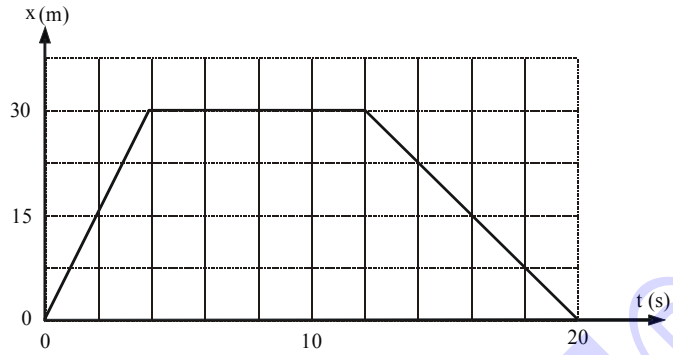
16. A particle moves along a straight line, x. At time t = 0, its position is at x = 0. The velocity, V, of the object changes as a function of time t, as indicated in the figure; t is in seconds, V in m/sec and x in meters.
(a) What is x at t = 3 sec?
(b) What is the instantaneous acceleration (in m/sec²) at t = 2 sec?
(c) What is the average velocity (in m/sec) between t = 0 and t = 3 sec?
(d) What is the average speed (in m/sec) between t = 1 and t = 3 sec?



KM0033

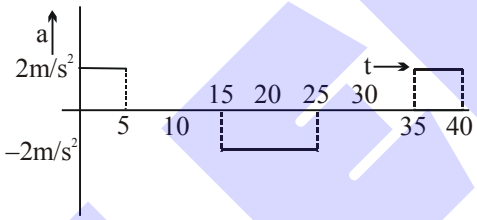
node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\1-U & D, Basic Maths & Vector, NUM, K-1D, 2D\Eng\02_Kinematics-1D & Calculus\00_Engs65

17. The graph illustrates motion of a bucket being lowered into a well from the top at the instant t = 0, down to the water level, filled with water and drawn up again. Here 'x' is the depth. Find the average speed of the bucket in m/s during whole operation.



KM0032

18. Figure shows a graph of acceleration of a particle moving on the x-axis. Plot the following graphs if the particle is at origin and at rest at t = 0.

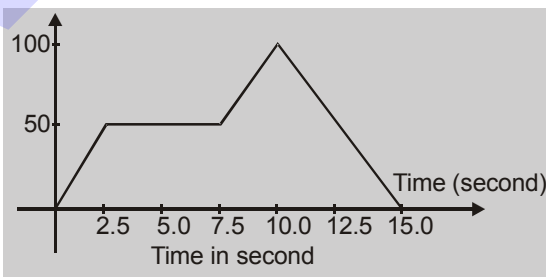


(i) velocity-time graph (ii) displacement-time graph (iii) distance-time graph. **KM0036**

19. A car, which can have constant acceleration of $4 m/s^2$ and a constant retardation of $8 m/s^2$ is travelling on a high way where maximum speed limit is 72 km/h. It starts froms rest and comes to rest in minimum time t_1 after travelling 300 meter and if there is no speed limit then minimum time taken is t_2, to cover the same distance. Then find the value of $\frac{4t_1}{t_2}$.

KM0312

20. Figure shows the graph of the x-coordinate of a particle going along the X-axis as a function of time. Find



(a) the average velocity during 0 to 10 s,
(b) instantaneous velocity at 2, 5, 8 and 12 s.

KM0313

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EXERCISE (JM)

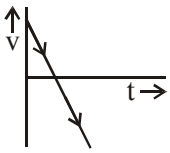
1. From a tower of height H, a particle is thrown vertically upwards with a speed u. The time taken by the particle, to hit the ground, is n times that taken by it to reach the highest point of its path. The relation between H, u and n is : **[JEE-Main-2014]**

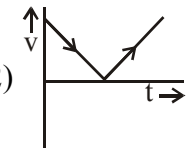
(1) $2gH = nu^2(n-2)$ (2) $gH = (n-2)u^2$ (3) $2gH = n^2u^2$ (4) $gH = (n-2)^2u^2$

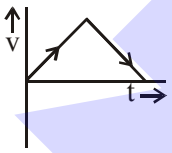
KM0104

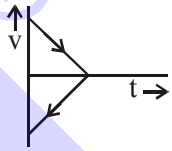
2. A body is thrown vertically upwards. Which one of the following graphs correctly represent the velocity vs time?

[JEE-Main-2017]

(1) 

(2) 

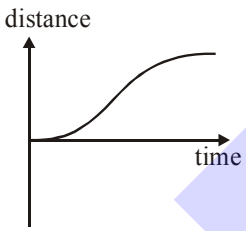
(3) 

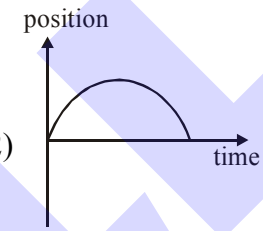
(4) 

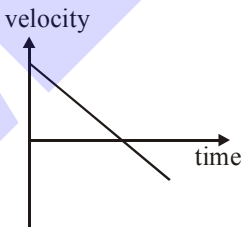
KM0105

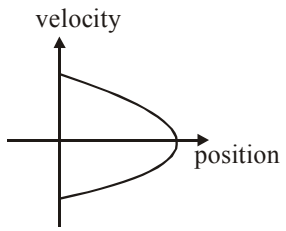
3. All the graphs below are intended to represent the same motion. One of them does it incorrectly. Pick it up.

[JEE-Main-2018]

(1) 

(2) 

(3) 

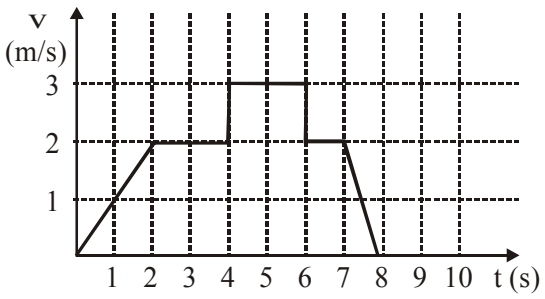
(4) 

KM0106

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

4. A particle starts from the origin at time t = 0 and moves along the positive x-axis. The graph of velocity with respect to time is shown in figure. What is the position of the particle at time t = 5s ?

[JEE-Main-2019_Jan]

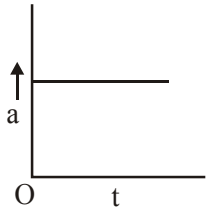


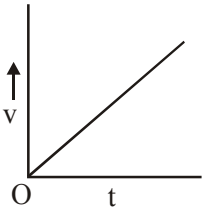
(1) 6 m (2) 9 m (3) 3 m (4) 10 m

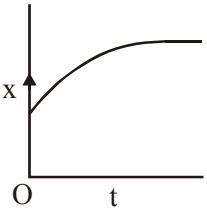
KM0314

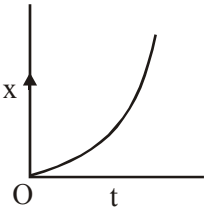
node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\1-U&D, Basic Maths & Vector, NUM, K-1D,2D\Eng\02_Kinematics-1D & Calculus\00_Engp65

5. A particle starts from origin O from rest and moves with a uniform acceleration along the positive x-axis. Identify all figures that correctly represent the motion qualitatively. (a = acceleration, v = velocity, x = displacement, t = time) **[JEE-Main-2019_April]**

(A) 

(B) 

(C) 

(D) 

(1) (A), (B), (C) (2) (A) (3) (A), (B), (D) (4) (B), (C)

KM0315

6. The position of a particle as a function of time t, is given by $x(t) = at + bt^2 - ct^3$; where a, b and c are constants. When the particle attains zero acceleration, then its velocity will be : **[JEE-Main-2019_April]**

(1) $a + \frac{b^2}{4c}$ (2) $a + \frac{b^2}{c}$ (3) $a + \frac{b^2}{2c}$ (4) $a + \frac{b^2}{3c}$

KM0316

7. A particle is moving with speed $v = b\sqrt{x}$ along positive x-axis. Calculate the speed of the particle at time $t = \tau$(assume that the particle is at origin at t = 0). **[JEE-Main-2019_April]**

(1) $\frac{b^2\tau}{4}$ (2) $\frac{b^2\tau}{2}$ (3) $b^2\tau$ (4) $\frac{b^2\tau}{\sqrt{2}}$

KM0317

8. The distance x covered by a particle in one dimensional motion varies with time t as $x^2 = at^2 + 2bt + c$. If the acceleration of the particle depends on x as x^{-n}, where n is an integer, the value of n is _____ . **[JEE-Main-2020_Jan]**

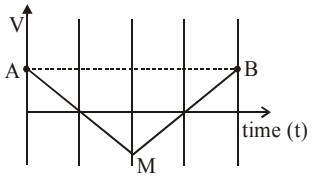
KM0318

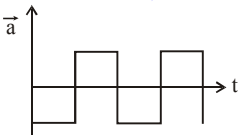
9. A helicopter reises from rest on the ground vertically upwards with a constant acceleration g. A food packet is dropped from the helicopter when it is a height h. The time taken by the packet to reach the ground is close to [g is the acceleration due to gravity] : **[JEE-Main-2020_Sep]**

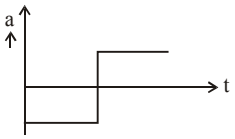
(1) $t = \sqrt{\frac{2h}{3g}}$ (2) $t = 1.8\sqrt{\frac{h}{g}}$ (3) $t = 3.4\sqrt{\left(\frac{h}{g}\right)}$ (4) $t = \frac{2}{3}\sqrt{\left(\frac{h}{g}\right)}$

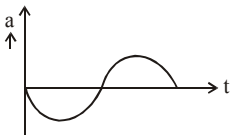
KM0319

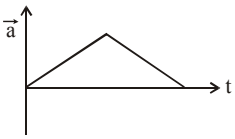
10. If the velocity-time graph has the shape AMB, what would be the shape of the corresponding acceleration-time graph ? **[JEE-Main-2021_Feb]**



(1) 

(2) 

(3) 

(4) 

KM0320

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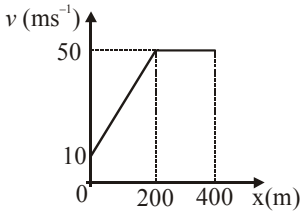
11. An engine of a train, moving with uniform acceleration, passes the signal-post with velocity u and the last compartment with velocity v. The velocity with which middle point of the train passes the signal post is:

[JEE-Main-2021_Feb]

(1) $\sqrt{\frac{v^2+u^2}{2}}$ (2) $\frac{v-u}{2}$ (3) $\frac{u+v}{2}$ (4) $\sqrt{\frac{v^2-u^2}{2}}$

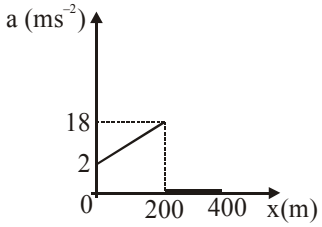
KM0321

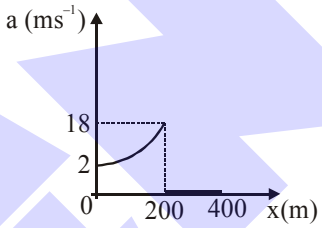
12. The velocity-displacement graph describing the motion of a bicycle is shown in the figure.

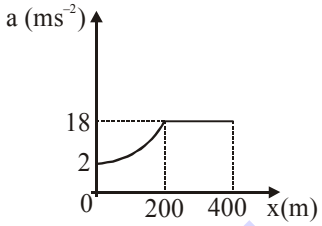


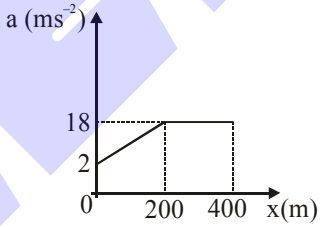
The acceleration-displacement graph of the bicycle's motion is best described by :

[JEE-Main-2021_March]

(1)


(2)


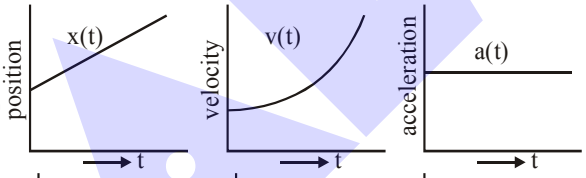
(3)


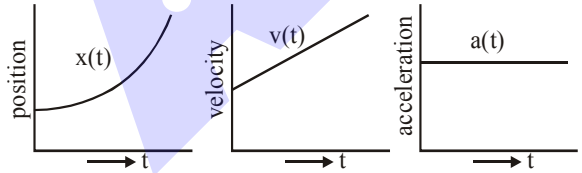
(4)


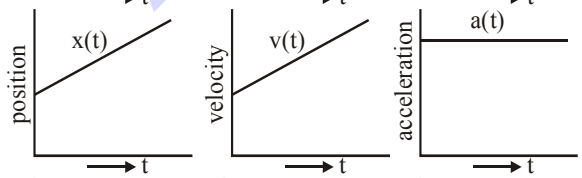
KM0322

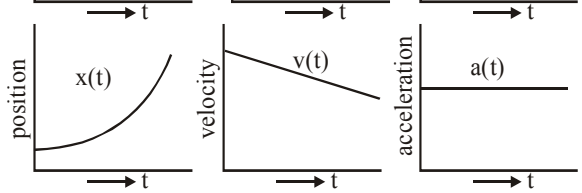
13. The position, velocity and acceleration of a particle moving with a constant acceleration can be represented by :

[JEE-Main-2021_March]

(1)


(2)


(3)


(4)


KM0323

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14. Water droplets are coming from an open tap at a particular rate. The spacing between a droplet observed at 4th second after its fall to the next droplet is 34.3 m. At what rate the droplets are coming from the tap ? (Take g = 9.8 m/s^2) **[JEE-Main-2021_July]**

(1) 3 drops / 2 seconds (2) 2 drops / second

(3) 1 drop / second (4) 1 drop / 7 seconds

KM0324

15. Two buses P and Q start from a point at the same time and move in a straight line and their positions are represented by $X_P(t) = \alpha t + \beta t^2$ and $X_Q(t) = ft - t^2$. At what time, both the buses have same velocity ? **[JEE-Main-2022_June]**

(1) $\frac{\alpha - f}{1+\beta}$ (2) $\frac{\alpha + f}{2(\beta - 1)}$ (3) $\frac{\alpha + f}{2(1+\beta)}$ (4) $\frac{f - \alpha}{2(1+\beta)}$

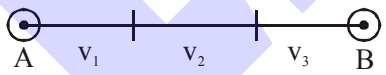
KM0325

16. An object is thrown vertically upwards. At its maximum height, which of the following quantity becomes zero ? **[JEE-Main-2022_June]**

(1) Momentum (2) Potential energy (3) Acceleration (4) Force

KM0326

17. A car covers AB distance with first one-third at velocity v_1 ms^{-1}, second one-third at v_2 ms^{-1} and last one-third at v_3 ms^{-1}. If $v_3 = 3v_1$, $v_2 = 2v_1$ and $v_1 = 11$ ms^{-1} then the average velocity of the car is _____ ms^{-1}. **[JEE-Main-2022_June]**



KM0327

18. A juggler throws balls vertically upwards with same initial velocity in air. When the first ball reaches its highest position, he throws the next ball. Assuming the juggler throws n balls per second, the maximum height the balls can reach is: **[JEE-Main-2022_July]**

(1) g/2n (2) g/n (3) 2gn (4) g/2n^2

KM0328

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EXERCISE (JA)

1. Consider an expanding sphere of instantaneous radius R whose total mass remains constant. The expansion is such that the *instantaneous* density ρ remains uniform throughout the volume. The rate of fractional change in density $\left(\frac{1}{\rho}\frac{d\rho}{dt}\right)$ is constant. The velocity v of any point on the surface of the expanding sphere is proportional to : **[JEE Advanced-2017]**

(A) R^3 (B) $\frac{1}{R}$ (C) R (D) $R^{2/3}$

KM0107

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ANSWER KEY

SUPPLEMENT FOR DIFFERENTIATION

1. $30x^5$ **2.** $3\cos x - 4\sin x$ **3.** $24x^3 - 3x^2$ **4.** $\frac{2}{\sqrt{x}} - 2$ **5.** $\frac{5}{x^2}$

6. $e^x + \frac{1}{x}$ **7.** $\frac{-2}{x^3} + 6\sqrt{x}$ **8.** $1 - 3\cos x$ **9.** **–3** **10.** (i) 5; (ii) 8.436

11. (i) $6x + 5$; (ii) 6; (iii) 0 **13.** $2x\sin x + x^2\cos x$ **14.** $e^x(\cos x - \sin x)$

15. $(2x + 3)\sin x + (x^2 + 3x - 4)\cos x$ **16.** $x(2\ln x + 1)$ **17.** $\cos^2 x - \sin^2 x$

18. $(x^2 + 2x)e^x + (2x + 2)e^x$ **19.** $\sin x + \frac{x}{\cos x}$ **20.** $\frac{5x^2 + 3}{2\sqrt{x}}$ **21.** $\sin 2x$

22. $\sec^2 x$ **23.** $\frac{1}{2}x^{-\frac{1}{2}} + \frac{3}{2}x^{\frac{1}{2}}$ **24.** $\sec x \tan x$ **26.** $e^x(\operatorname{cosec} x - \operatorname{cosec} x \cot x)$

27. $e^x\left(\frac{x-1}{x^2+1}\right)^2$ **28.** 8/9 **29.** $4\sqrt{3}$ **30.** $8(2x + 3)^3$ **31.** $3\sin^2 x \cos x$

32. $4e^{4x+3}$ **33.** $6x(x^2 + 2)^2$ **34.** $= \frac{6x+1}{3x^2+x}$ **35.** $4\sin(8x + 10)$ **36.** $\cos x \sin x$

37. $\frac{3}{4}$ **38.** $\cot x$ **39.** $\sec x$ **40.** 3 **41.** 7

42. 5 **43.** $\sqrt{2}$ **44.** $y_{max} = \frac{16}{3}$ and $y_{min} = -\frac{16}{3}$ **45.** (D)

SUPPLEMENT FOR INTEGRATION

1. $2x(\sqrt{x} + 1) + C$ **2.** $x^3 + x + 3x + C$ **3.** $\frac{2}{7}z^{7/2} - \frac{1}{2}z^{1/2} + C$ **5.** $\frac{2}{3}\sqrt{x}(x + 1)$ **6.** $8\ln x + \frac{5}{x} - \frac{3}{x^2} + C$

7. $x - \frac{4}{x} + C$ **8.** $\tan\theta$ **9.** $\theta + \sin\theta + C$ **10.** $\theta - \sin\theta + C$ **11.** $-3\cos 3x + C$

12. $3\sin 4\theta + C$ **13.** $\frac{-4}{7}e^{-7x} + C$ **14.** $\frac{-5\sin \pi x}{\pi} + C$ **15.** $x^3 + 2e^{2x+3} + C$ **16.** $\ln(4x + 3) + C$

17. $\frac{\ln(4x+3) + \ln(4x+5)}{4} + C$ **18.** $\frac{1}{5}\sin(5x - 3) + \frac{1}{2}\sin(2x + 5) + C$ **19.** $\sqrt{3}\tan^{-1}(x) + C$

20. $\frac{-1}{3(4x+3x^2)} + C$ **21.** $\ln(\sec x)$ **22.** $\ln(\sec x + \tan x) + C$ **23.** $\frac{\sin^5 x}{5} + C$

24. $\sin x - \frac{\sin^3 x}{3} + C$ **25.** $\frac{\cos^3 x}{3} - \cos x + C$ **26.** $\frac{2\pi + 3}{6}$ **27.** 60 **28.** 200.2

29. 14.25 **30.** $2(e^2 - 1)$ **31.** $-\frac{\sqrt{2}}{6}$ **32.** $\ln 9$ **33.** 3 **34.** 3

35. $-\frac{\sqrt{3}}{8}$ **36.** $\ln(\ln 3)$ **37.** (a) $\frac{dy}{dx} = 4x\cos 2x$; (b) $\frac{1}{8}(\pi - 2)$ **38.** (a) 2; (b) 7.3 **39.** **1** **40.** $\frac{1}{4\sqrt{2}}$

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EXERCISE # O

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	B	A	B	B	A	C	A	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	B	B	B	D	C	C	C	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	A	D	B	B	D	B	B	D	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	D	A	A	B	C	A	A	B,C,D	C,D
Que.	41	42	43	44	45	46	47	48	49	
Ans.	A,B,D	A,C	A,D	A,C,D	A,B	B	D	A-R,T; B-T; C-R,T; D-S,T	A-R; B-P; C-S; D-Q	

EXERCISE # S

1. (a) 100m; (b) 10 s; (c) 19m, 15m **2.** (a) $2\sqrt{2}\frac{v}{\pi}$, (b) $\frac{2v}{\pi}$, (c) $\frac{2\sqrt{2}v}{3\pi}$ **3.** 18.5 m

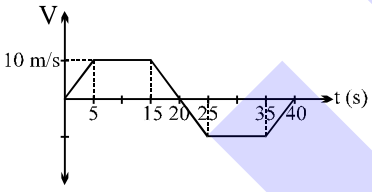
4. 5h/3 **5.** 6 **6.** (a) 27 m/s, (b) 108 m, (c) –100 m, (d) 316 m

7. 1 kg m/sec. **8.** 186.25 N **9.** 23 amp **10.** –72, –16, 24

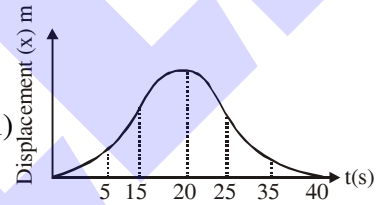
11. (a) 0, 8 sec (b) 0, 4, 8 sec **12.** 1.815×10^5 ergs. **13.** 100 kgm/s **14.** 3

15. $\left(27\hat{i}+2\hat{k}\right)m/s^2$ **16.** (a) 3 m; (b) –3 m/s^2; (c) 1m/s; (d) 3/2 /s **17.** 21 m/s

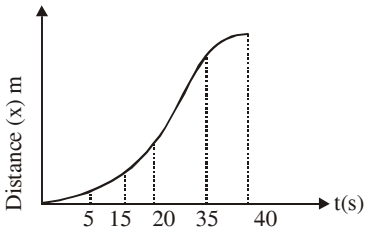
18. (i)



(ii)



(iii)



19. 5 **20.** (a) 10 m/s, (b) 20 m/s, zero, 20 m/s, –20 m/s

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	1	1	2	3	4	2	3	3	2
Que.	11	12	13	14	15	16	17	18		
Ans.	1	1	2	3	4	1	18	4		

EXERCISE # JEE-ADVANCED

Que.	1
Ans.	C

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